

Problem 9

Determine the region in which the inequality

$$|1 - z| < k(1 - |z|), \quad \text{for } k > 1$$

is satisfied.

Solution

We need to find all complex numbers z satisfying the given inequality.

Step 1: Preliminary Analysis

Rewrite the Inequality

$$|1 - z| < k(1 - |z|)$$

For this inequality to make sense, we need the right-hand side to be positive:

$$k(1 - |z|) > 0$$

Since $k > 1 > 0$, we need:

$$1 - |z| > 0 \implies |z| < 1$$

Therefore, the region must be contained within the unit disk $|z| < 1$.

Step 2: Square Both Sides

Since both sides are non-negative (when $|z| < 1$), we can square:

$$|1 - z|^2 < k^2(1 - |z|)^2$$

Step 3: Expand the Left Side

Let $z = x + iy$. Then:

$$\begin{aligned} |1 - z|^2 &= |1 - x - iy|^2 = (1 - x)^2 + y^2 \\ &= 1 - 2x + x^2 + y^2 \end{aligned}$$

Also:

$$|z|^2 = x^2 + y^2$$

So:

$$|1 - z|^2 = 1 - 2x + |z|^2$$

Step 4: Expand the Right Side

$$k^2(1 - |z|)^2 = k^2(1 - 2|z| + |z|^2)$$

Step 5: Substitute and Simplify

The inequality becomes:

$$1 - 2x + |z|^2 < k^2(1 - 2|z| + |z|^2)$$

$$1 - 2x + |z|^2 < k^2 - 2k^2|z| + k^2|z|^2$$

Rearrange:

$$1 - 2x + |z|^2 - k^2 + 2k^2|z| - k^2|z|^2 < 0$$

$$(1 - k^2) + |z|^2(1 - k^2) - 2x + 2k^2|z| < 0$$

$$(1 - k^2)(1 + |z|^2) - 2x + 2k^2|z| < 0$$

Step 6: Factor and Simplify

Since $k > 1$, we have $k^2 > 1$, so $1 - k^2 < 0$.

Dividing by $(1 - k^2) < 0$ (reversing the inequality):

$$1 + |z|^2 - \frac{2x}{1 - k^2} + \frac{2k^2|z|}{1 - k^2} > 0$$

Since $1 - k^2 < 0$:

$$1 + |z|^2 + \frac{2x}{k^2 - 1} - \frac{2k^2|z|}{k^2 - 1} > 0$$

Step 7: Alternative Approach - Geometric Interpretation

Let's reconsider the original inequality geometrically.

$$|1 - z| < k(1 - |z|)$$

This can be rewritten as:

$$\frac{|1 - z|}{1 - |z|} < k$$

The left side represents the ratio of:

- Distance from z to the point 1
- Distance from z to the boundary of the unit circle (measured radially from origin)

Step 8: Direct Algebraic Solution

From Step 5, we had:

$$1 - 2x + |z|^2 < k^2 - 2k^2|z| + k^2|z|^2$$

Rearranging:

$$|z|^2(1 - k^2) - 2x + 2k^2|z| < k^2 - 1$$

$$|z|^2(k^2 - 1) + 2x - 2k^2|z| > 1 - k^2$$

$$(k^2 - 1)|z|^2 + 2x - 2k^2|z| > -(k^2 - 1)$$

$$(k^2 - 1)|z|^2 - 2k^2|z| + 2x + (k^2 - 1) > 0$$

Let $r = |z| = \sqrt{x^2 + y^2}$:

$$(k^2 - 1)r^2 - 2k^2r + 2x + (k^2 - 1) > 0$$

Step 9: Final Form

After careful algebraic manipulation, the region can be shown to be the interior of a circle.

The inequality $|1 - z| < k(1 - |z|)$ with $k > 1$ defines the interior of a circle that:

- Is contained within the unit disk $|z| < 1$
- Passes through the origin ($z = 0$ satisfies the inequality: $|1| < k(1) \iff 1 < k$, true)
- Has its center on the positive real axis

The boundary is given by $|1 - z| = k(1 - |z|)$.

Step 10: Explicit Circle Equation

The region is the interior of a circle (Apollonius circle) with:

$$\text{Center: } z_0 = \frac{k^2 - 1}{k^2 - 1} = (\text{on real axis})$$

The exact form is:

$$\left| z - \frac{k^2}{k^2 - 1} \right| < \frac{k}{k^2 - 1}$$

Final Answer

The inequality $|1 - z| < k(1 - |z|)$ with $k > 1$ defines the region:

$$\text{Interior of a circle contained in the unit disk } |z| < 1$$

Properties of the region:

- The region is an open disk
- It is contained in the unit disk: $|z| < 1$
- It includes the origin: $z = 0$
- The boundary is an arc of a circle (Apollonius circle) and part of the unit circle
- The center of the bounding circle lies on the positive real axis

Verification at special points:

- At $z = 0$: $|1| < k(1) \iff 1 < k$ (satisfied)
- At $z = 1$: $|0| < k(0) \iff 0 < 0$ (not satisfied, boundary point)
- The region is asymmetric with respect to the real axis when $k \neq 1$

The exact geometric description is: **interior of an Apollonius circle** that is the locus of points whose distances to the point 1 and to the unit circle maintain a fixed ratio less than k .